INCH-POUND
MIL-DTL-24643/72
9 October 2008

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO + 90 °C, 1000 VOLTS, TYPE LSFCF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

Components 1- 4:

First	-	Copper conductor, tin-coated, 4AWG 420 strands of 30AWG.
Second	-	Flexible silicone insulation. Diameter: 0.380±0.010 inch
Third	-	Insulation covering - flexible thermoset insulation. Color black. Diameter: 0.450±0.010 inch. Printed white identification marked at a maximum distance of 3 inches, 1-Black, 2-White, 3-Red, 4-Green.

Components 5-8:

First	-	Copper conductor, tin-coated, 16AWG 19 strands of 29AWG.
Second	-	Thermoset insulation. Color black. Diameter: 0.090±0.003 inch. Printed white identification marked at a maximum distance of 3 inches, 5-Orange, 6-Blue, 7-White-Black, 8-Red-Black.

Assembly:

Fourth	-	The components 1 – 4 and 5 – 8 shall be cabled together with a maximum lay of 11 inches. Component 5 shall be placed in the outer interstices between component 1 and 2. Component 6 shall be placed in the outer interstices between component 2 and 3. Component 7 shall be placed in the outer interstices between component 3 and 4. Component 8 shall be placed in the outer interstices between component 4 and 1. Fillers may be used as necessary to form a firm well-rounded assembly.
Fifth	-	Binder tape applied helically with overlap.
Sixth	-	Braided shield of AWG 34 tin-coated copper. Braid angle of 40 to 50 degrees; minimum coverage of 90 percent.
Seventh	-	Optional binder tape applied helically with overlap.
Eighth	-	Flexible cross-linked polyolefin jacket with a nominal wall thickness of 0.075 inch. Color black.

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TABLE I. Construction details.

Military Part No. M24643/72	Type & Size	Minimum Bend Radius (inch)	Overall Diameter (inches)	
			Type LSFSU	
			(min.)	(max.)
-01UO	LSFCF-42	7.5	1.250	1.350

REQUIREMENTS:

Qualification required.

EXAMINATION AND TESTS:

Basic Electricals:

Conductor resistance (4AWG) (ohms/1000 feet at 25 °C, max.)	0.298
Conductor resistance (16AWG) (ohms/1000 feet at 25 °C, max.)	4.62
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	3000
Conductor to shield	3000
Insulation resistance (megohms/1000 feet, min.)	See table I
Conductor to conductor	500
Conductor to shield	250
Jacket flaws	No Failure
Conductor and shield continuity	No Failure
Group A:	
Visual and dimensional	No Failure
Group B:	
Cross-linked proof test (percent, max.)	
Insulation	50
Jacket	50
Cold bending, cable mandrel diameter 12 times cable diameter	No Damage
Bending endurance (25±2 °C, cycles, min.)	25
Physicals (unaged)	
Insulation (flexible silicone 4AWG and thermoset insulation 16AWG)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	250

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Insulation covering (4AWG only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	200
Durometer (Type A hardness, min./max.)	60/75
Jacket (overall cable)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	200
Durometer (Type A hardness, min./max.)	60/75
Tear (pounds per inch thickness, min.)	35
Group C:	
Physicals (aged)	
Air oven	
Insulation (inner layer)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Insulation covering (4AWG only)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Jacket (overall cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Heat distortion (percent of unaged, max.)	30
Permanence of printing (insulation) (cycles, min.)	75
Shrinkage	No Failure
Shield - conformance to material (construction and coverage)	No Failure
Group D:	
Flame propagation (cable)	No Failure
Qualification Inspection:	
Qualification inspection shall include basic electricals, groups A, B, C, and D, plus the following:	
Aging and compatibility (cable) (125±5 °C)	No Failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2

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Fillers	2
Insulation	2
Halogen content (percent, max.)	
Insulation	0.2
Jacket	0.2
Fillers	0.2
Smoke index, max.	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, max.	
Jacket	5
Fillers	5
Insulation	1.5
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Durometer (jacket – Type A hardness, min.)	80
Electrical moisture absorption	No Failure
UNIT ORDERING LENGTHS: 250 feet (nominal)	

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-119)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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